



Taylor's series



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$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots + \frac{h^n}{n!} f^n(x) + \dots \quad \text{-----(1)}$$

This is known as **Taylor's series**.

interchanging x and h in (1)

$$f(x+h) = f(h) + xf'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \dots + \frac{x^n}{n!} f^n(h) + \dots \quad \text{-----(2)}$$

replace x by a & h by $(x-a)$ in (1)

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots + \frac{(x-a)^n}{n!} f^n(a) + \dots \quad \text{-----(3)}$$

$$\sinh(x+a) = \sinh a + x \cosh a + \frac{x^2}{2!} \sinh a + \dots$$



1) Given that $\sinh(1.5) = 2.1293$ & $\cosh(1.5) = 2.3524$

calculate $\sinh(1.505)$.

Solution: $f(x+h) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \frac{x^3}{3!}f'''(h) + \dots$

$$h = 1.5 \quad x = 0.005$$

$$\sinh 1.505 = \sinh(1.5) + (0.005) \cosh(1.5) + \frac{(0.005)^2}{2!} \sinh(1.5) + \dots$$

$$= 2.1293 + 0.005(2.3524) + (0.000025)(2.1293)$$

$$= 2.1411$$



2) Using Taylors theorem evaluate $\sqrt{1.02}$

Solution: $x = 1$ $h = 0.02$

$$f(x) = \sqrt{x} \quad f(1) = 1$$

$$f'(x) = \frac{1}{2\sqrt{x}} \quad f'(1) = \frac{1}{2}$$

$$f''(x) = \frac{1}{4}x^{-3/2} \quad f''(1) = -\frac{1}{4}$$

$$f'''(x) = \frac{3}{8}x^{-5/2} \quad f'''(1) = \frac{3}{8}$$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots$$

$$f(1.02) = f(1) + (0.02)f'(1) + \frac{(0.02)^2}{2!}f''(1) + \frac{(0.02)^3}{3!}f'''(1) + \dots$$

$$\sqrt{1.02} = 1 + (0.02)\frac{1}{2} + \frac{(0.02)^2}{2!}\left(-\frac{1}{4}\right) + \frac{(0.02)^3}{3!}\left(\frac{3}{8}\right) + \dots = 1.0099$$



3) Expand $\tan\left(\frac{\pi}{4} + x\right)$ and hence find $\tan(46^\circ, 30')$.

solution: $f(x) = \tan x$ $f\left(\frac{\pi}{4}\right) = 1$

$$f'(x) = \sec^2 x = 1 + \tan^2 x = 1 + f^2 \quad f'\left(\frac{\pi}{4}\right) = 2$$

$$f''(x) = 2ff' \quad f''\left(\frac{\pi}{4}\right) = 2 \times 1 \times 2 = 4$$

$$f'''(x) = 2[ff'' + f'^2] \quad f'''\left(\frac{\pi}{4}\right) = 2(1 \times 4 + 4) = 16$$

$$f^4(x) = 2[ff^3 + f'f'' + 2f'f''] \quad f^4\left(\frac{\pi}{4}\right) = 2(1 \times 16 + 2 \times 4 + 2 \times 2 \times 4) = 2[16 + 8 + 16] = 80$$

$$f\left(\frac{\pi}{4} + x\right) = f\left(\frac{\pi}{4}\right) + xf'\left(\frac{\pi}{4}\right) + \frac{x^2}{2!}f''\left(\frac{\pi}{4}\right) + \frac{x^3}{3!}f'''\left(\frac{\pi}{4}\right) + \frac{x^4}{4!}f^4\left(\frac{\pi}{4}\right) + \dots$$

$$= 1 + x \cdot 2 + \frac{x^2}{2!} \cdot 4 + \frac{x^3}{3!} \cdot 16 + \frac{x^4}{4!} \cdot 80 + \dots$$

$$= 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \frac{10}{3}x^4 + \dots$$



$$f(46^\circ, 30') = f(45^\circ + 1^\circ 30')$$

$$= 1 + 2 \times \frac{\pi}{120} + 2 \times \left(\frac{\pi}{120}\right)^2 + \frac{8}{3} \times \left(\frac{\pi}{120}\right)^3 + \frac{10}{3} \times \left(\frac{\pi}{120}\right)^4 + \dots$$

$$= 1.0574$$

4) Find $\sqrt{9.12}$.

(ans:3.0199)



5) If $f(x) = x^3 + 3x^2 + 15x - 10$ Find $f(1.1)$

Soln: Here $x = 1$ $h = 0.1$

$$f(x) = x^3 + 3x^2 + 15x - 10 \quad f(1) = 9$$

$$f'(x) = 3x^2 + 6x + 15 \quad f'(1) = 24$$

$$f''(x) = 6x + 6 \quad f''(1) = 12$$

$$f'''(x) = 6 \quad f'''(1) = 6$$

$$f^4(x) = 0 \quad f^4(1) = 0$$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots$$

$$= f(1) + (0.1)f'(1) + \frac{(0.1)^2}{2!}f''(1) + \frac{(0.1)^3}{3!}f'''(1) + 0$$

$$= 9 + (0.1) \times 24 + \frac{(0.1)^2}{2!} \times 12 + \frac{(0.1)^3}{3!} \times 6 = 11.461$$



6) Expand $f(x) = x^4 - 3x^3 + 2x^2 - x + 1$ in powers of $(x - 3)$ using Taylor's Theorem.

Soln: $f(x) = x^4 - 3x^3 + 2x^2 - x + 1$ $f(3) = 16$

$$f'(x) = 4x^3 - 9x^2 + 4x - 1 \quad f'(3) = 38$$

$$f''(x) = 12x^2 - 18x + 4 \quad f''(3) = 58$$

$$f'''(x) = 24x - 18 \quad f'''(3) = 54$$

$$f''''(x) = 24 \quad f''''(3) = 24$$

$$f(x) = f(a) + (x - a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots + \frac{(x-a)^n}{n!} f^n(a) + \dots$$

Put $a = 3$

$$f(x) = f(3) + (x - 3)f'(3) + \frac{(x - 3)^2}{2!} f''(3) + \frac{(x - 3)^3}{3!} f'''(3) + \frac{(x - 3)^4}{4!} f''''(3) + \dots$$

$$= 16 + 38(x - 3) + 58 \frac{(x - 3)^2}{2!} + 54 \frac{(x - 3)^3}{3!} + 24 \frac{(x - 3)^4}{4!} + \dots$$

$$= 16 + 38(x - 3) + 29(x - 3)^2 + 9(x - 3)^3 + (x - 3)^4 + \dots$$



7) Expand $f(x) = x^5 - 5x^4 + 6x^3 - 7x^2 + 8x - 9$ in powers of $(x - 1)$.

ans: $-6 - 3(x - 1) - 9(x - 1)^2 - 4(x - 1)^3 + (x - 1)^5$